

INPUT

Origin-destination flows
of freight good

Electricity prices

Technical & Cost parameters of
trucks

Payload, specific energy
consumption, battery/ tank size |
Vehicle purchase | O&M costs |

Grid connection costs

Technical parameters of trucks
generation dependent technical
performance related to driving range
and charging power

Parameters on charging
infrastructure expansion

Differentiated by charging power &
utilization

Mode shift to rail

Available capacities in rail
transport | transport costs of rail |
infrastructure expansion costs

Optimization model

- **Linear program**
- **based on Network Design Problem**
- System cost-minimization
- Geographic resolution: NUTS-2
- Temporal resolution: annual
- Network-based transport demand coverage
- Technology & mode shift
- Sizing and allocation of charging infrastructure

OUTPUT

Trips per mode & technology

Vehicle stock shift

Charging demand by location

Charging infrastructure expansion

Mode split